

QXU5030 Composite Materials

2025B | Prof. Shafique Ahmed

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1. Introduction to Module

■ Assessment

Activity	Score
Coursework(labs)	40%
Final Exam	60%

Week 2N+1: Kahoot! quiz with immediate flash feedback provided after each question.
 Week 2N: Quiz delivered via QM+ (or paper-based assignment), with structured written feedback provided within one week.

1.1. Basic Concepts

Anisotropy

A property of a material show different mechanical or physical properties in different directions.

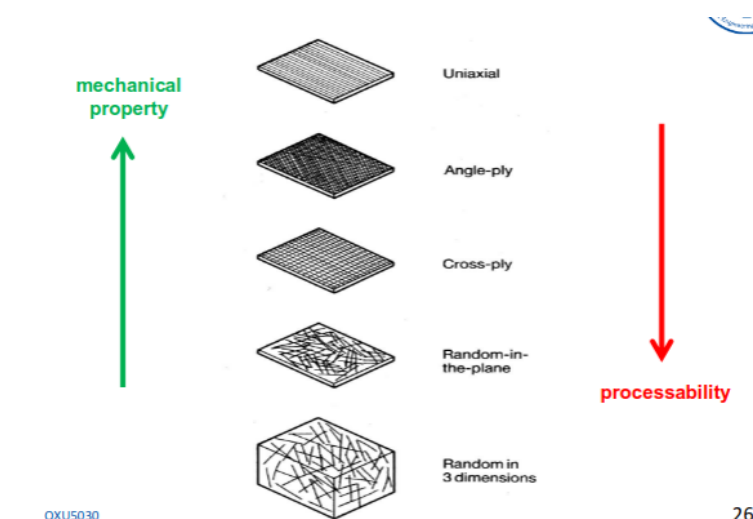
■ Man-Made Composite

Polymer Matrix Composites (PMC's) These are the most common and will be the main area of discussion in this course. Also known as FRP - Fibre Reinforced Polymers (or Plastics) - these materials use a polymer-based resin as the matrix, and a variety of fibres such as glass, carbon and aramid as the reinforcement.

Metal Matrix Composites (MMC's) - Increasingly found in the automotive industry, these materials use a metal such as aluminium as the matrix, and reinforce it with fibres such as silicon carbide.

Ceramic Matrix Composites (CMC's) - Used in very high temperature environments, these materials use a ceramic as the matrix and reinforce it with short fibres, or whiskers such as those made from silicon carbide and boron nitride.

■ Type of Composites



1.2. Properties of Composites

■ Factors of Composites Properties

The properties of the composite are determined by:

1. The properties of the fibre
2. The properties of the resin
3. The ratio of fibre to resin (Fibre Volume Fraction)
4. The geometry and orientation of the fibres

■ Specific Strength

Strength-to-weight ratio = tensile strength/density

Indicates how strong a material is relative to its weight.

■ Lightness, or E/ρ^3 and Lightweight Structural Design

E = Young's modulus (stiffness)

E/ρ^3 = key parameter for bending stiffness and buckling resistance in weight-sensitive structures

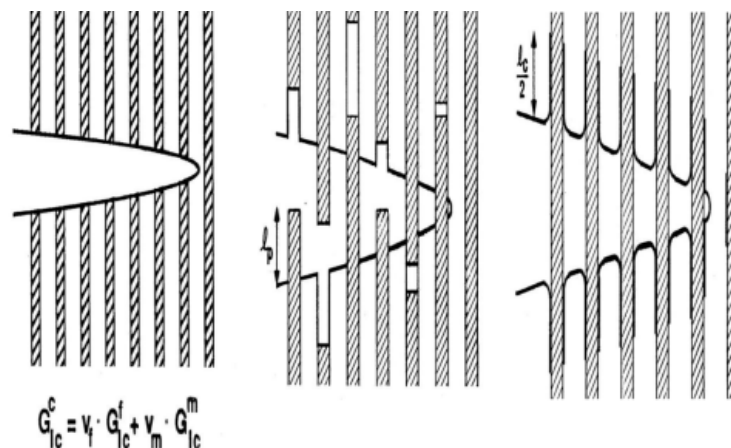
■ Synergism, the "Composite Effect"

Ideally, rule of mixture (RoM) for a physical property P :

$$P_{\text{RoM}} = P_1 V_1 + P_2 V_2 \quad (V_1 + V_2 = 1)$$

However, synergism can make $P_{\text{observed}} \gg P_{\text{RoM}}$

And synergism can modify the mode of fracture by weakening of fibre/matrix interface to prevent brittle fracture.



■ Safety

Composites can absorb more energy per kilo than metals in crash.